

*— Adam Smith, **The Wealth of Nations** · 1776*

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Amazon Bedrock AgentCore in Production: What Nobody Tells You

Join the CoE

We build projects like the one you'll see today



- I work with microservices, serverless, and cloud-native systems on AWS and Azure.
- Currently - Agentic AI with Amazon Bedrock and Python.
- +8 years of teaching passion with +1,200 students trained.



Jhon Robert Quintero

Technical Architect

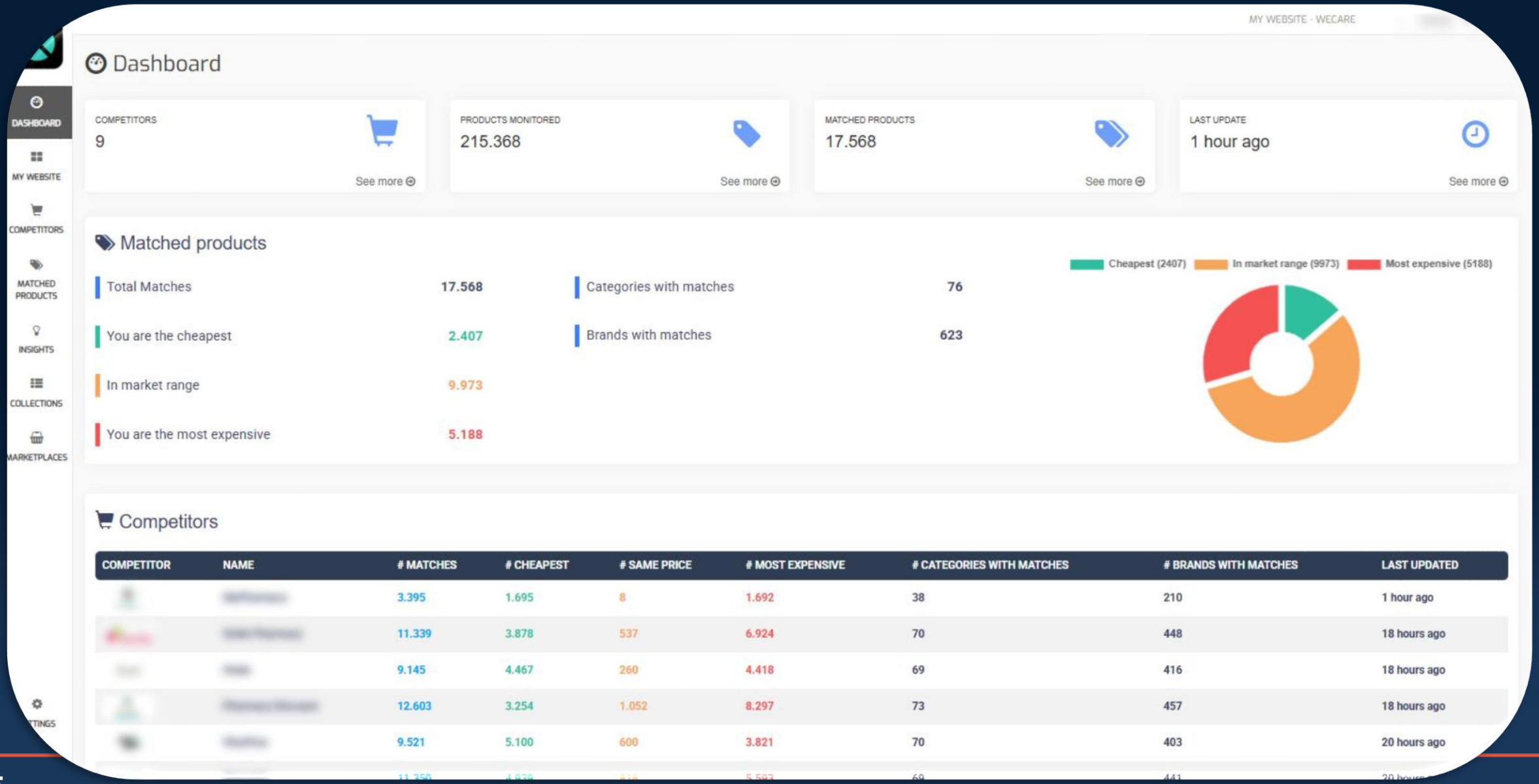
Agenda

01 *I introduce the character and the situation*

02 *The conflict or the change happens*

03 *It resolves with a lesson or takeaway*

The Challenge



How to collect information from thousands of products daily,

automatically, reliably, and without blowing the budget?



Automatic
No manual intervention

Reliable
Accurate and consistent data

Scalable
Thousands of products, every day

Cost-effective
Maximum value, minimum cost

Why is it hard?

The 3 enemies: Volume, Intelligence, Cost

01

Volume & Speed

Thousands of products across multiple platforms, updated in real time, requiring continuous ingestion processes.

02

Intelligence Layer

Raw HTML required AI-based classification, data extraction, and deduplication with no human review.

03



Cost at Scale

Every time a foundation model is invoked, a cost is incurred. Multiply that by thousands of daily records and costs become considerable.

The architecture we chose

1. SOURCES

E-commerce Sites

- amazon.com
- mercadolibre.com
- walmart.com
- ebay.com
- bestbuy.com
- target.com

...

Thousands of products across hundreds of sites

2. ORCHESTRATION



Amazon EventBridge
Triggers the daily collection



AWS Step Functions
Orchestrates and distributes the tasks

Dynamic configuration

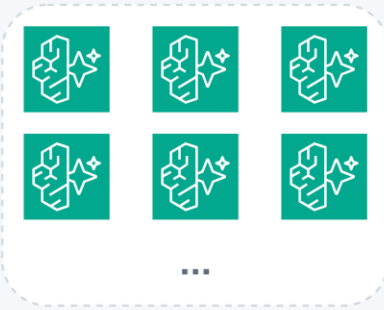
- Sites
- Categories
- Frequency
- CSS/XPath selectors
- Business rules

3. PARALLEL EXECUTION

Amazon Bedrock AgentCore

Hundreds or thousands of agents collecting in parallel



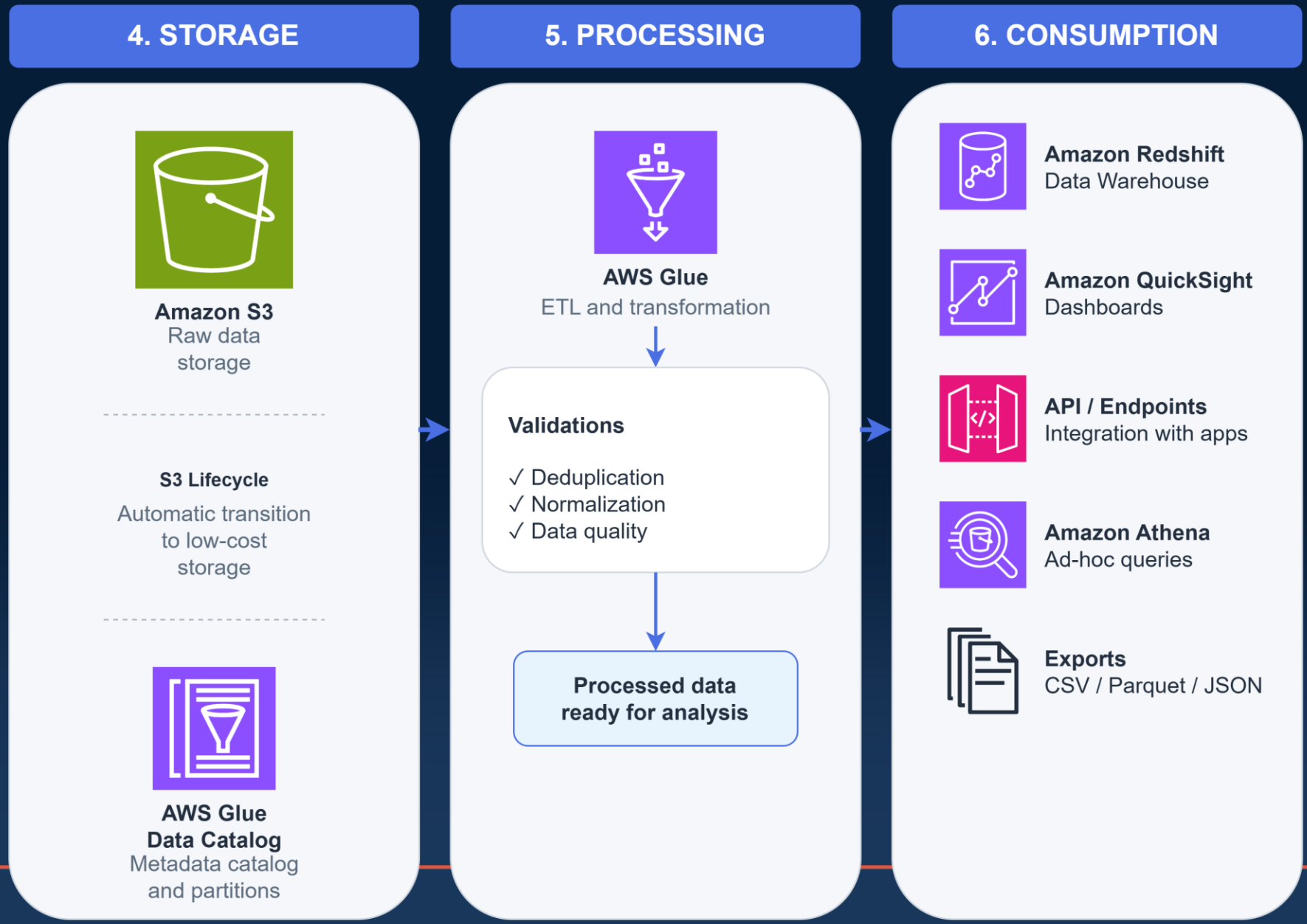


...

Each agent extracts:

- Product information
- Price
- Availability
- Reviews
- Images
- Specifications

The architecture we chose



The Surprises

What we discovered in production

01

15-min Timeout

AgentCore cuts execution at 15 minutes. We solved it by wrapping the handler in **`@app.async_task`** so the runtime tracks busy/healthy status with Strands.

02

Context Overflow

Large HTML pages exceeded the context and caused silent truncation. Solution: **chunking** + recursive summarization, critical fields at the start of the prompt, and token monitoring with CloudWatch.

03

No Context Carryover

In parallel processing, reusing the conversation contaminates extractions. Each agent is created **from scratch, with no history**.

STRANDS AGENTS SDK





The great model dilemma

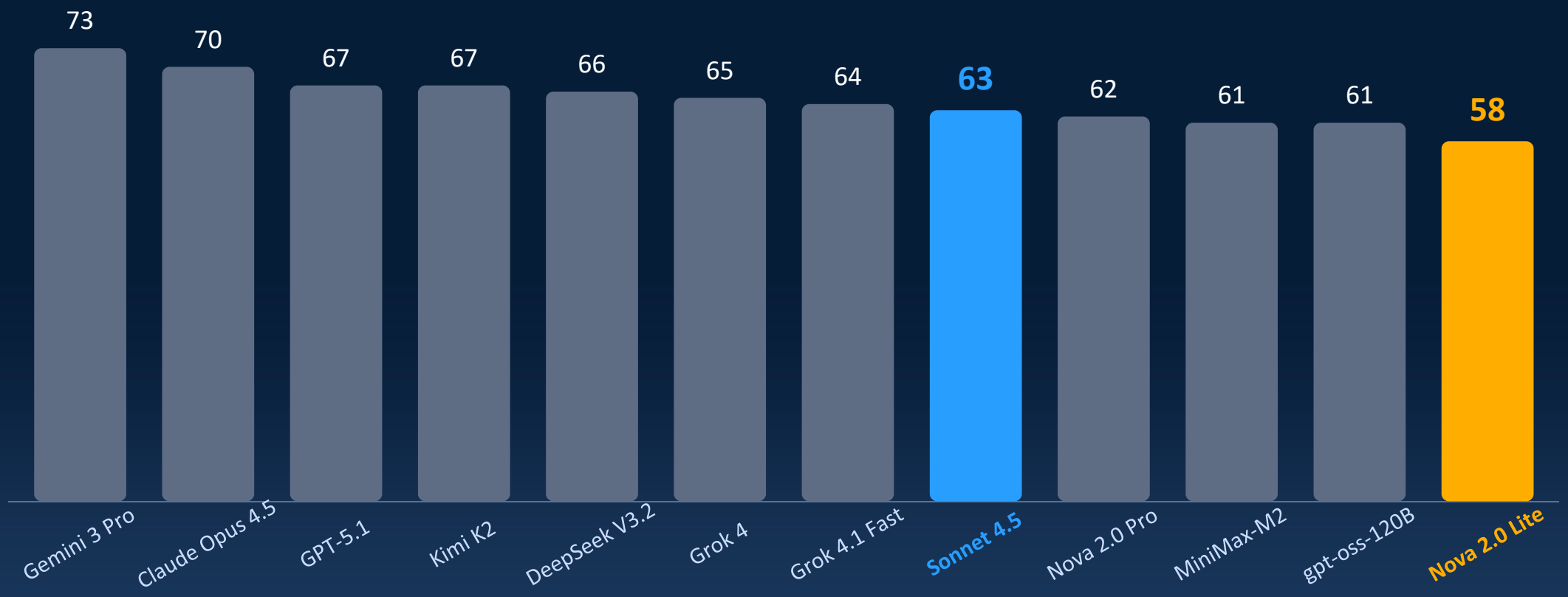
The question that changed the system's economics

> **Are we paying for the quality we really need?**

Before choosing a model, you must know what quality level your use case really demands.

What the market says

Artificial Analysis Intelligence Index v3.0 (December 2025)



Just 5 points separate Sonnet 4.5 from Nova 2.0 Lite. The question is: how much do you pay for those 5 points?

Nova 2 Lite vs Sonnet 4.5

Systematic comparison on a dataset

Metric	Nova Lite	Nova 2 Lite	Sonnet 4.5
Product Coverage	55%	**97% ★**	94%
Field Preservation	97%	99%	**100% ★**
HTML Data Loss	16 fields	7 fields	**0 fields ★**
Bonus Fields Added	402	654	**964 ★**
Schema Errors	2	1	**0 ★**
Data Quality Warnings	3	0	**0 ★**

★ winner in the metric | Dataset: product extraction on real e-commerce sites

The numbers speak

Winner by category

PRODUCT COVERAGE	FIELD PRESERVATION	DATA ENRICHMENT	SCHEMA COMPLIANCE
97%	**100%**	**+964**	**0**
Nova 2 Lite	**Sonnet 4.5**	**Sonnet 4.5**	**Sonnet 4.5**
Only 6 products missed	*Zero HTML data loss*	*Extra fields extracted*	*JSON with no errors*

Sonnet wins on quality. Nova 2 Lite wins on coverage.

The decision and why

Weighted ranking and layered model strategy

01

Nova 2 Lite

"The Reliable Workhorse"

WEIGHTED SCORE

93.5

BEST FOR

Maximum data capture with good fidelity

02

Sonnet 4.5

"The Quality Leader"

WEIGHTED SCORE

93.0

BEST FOR

Maximum data quality and field extraction

03

Nova Lite

"Needs Improvement"

WEIGHTED SCORE

73.5

BEST FOR

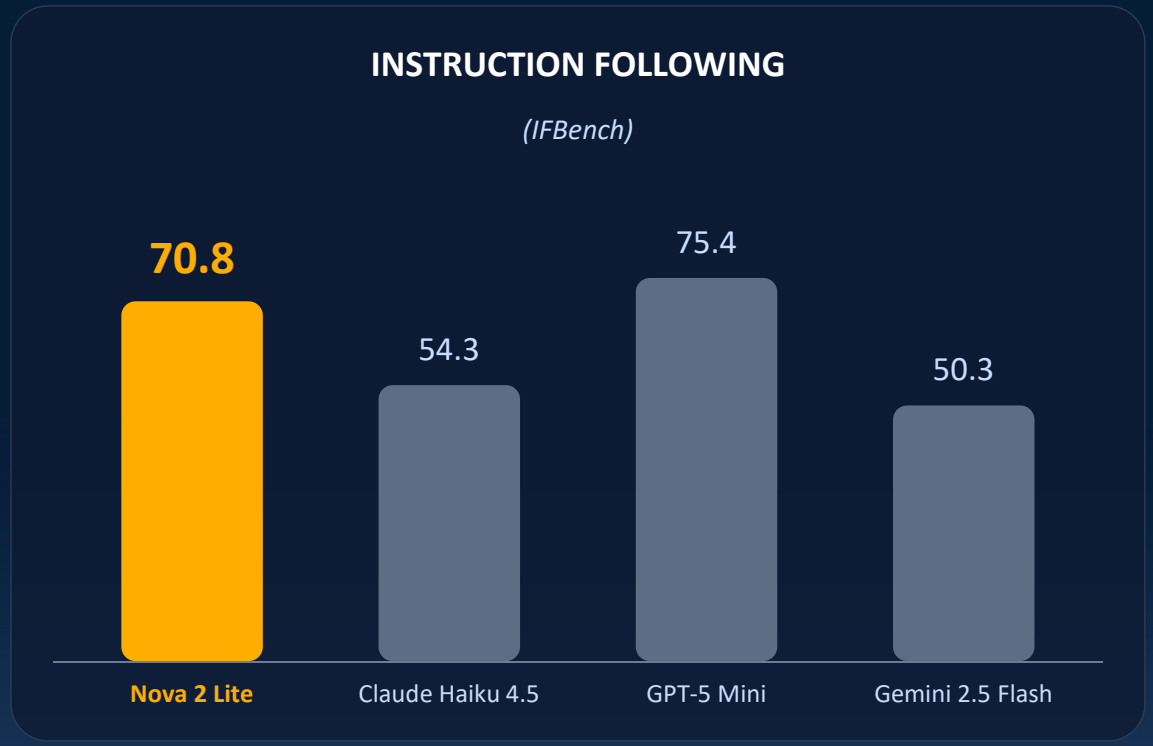
Not recommended as primary model

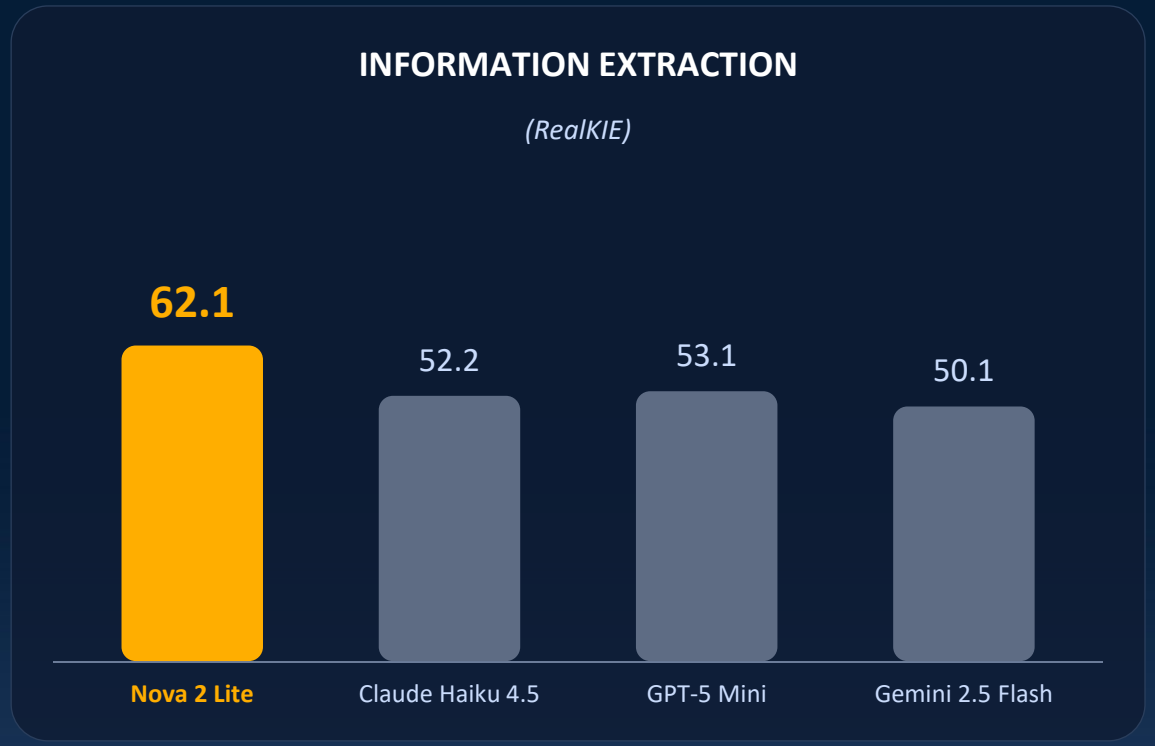
OUR DECISION

Nova 2 Lite as primary (coverage + cost) + Sonnet 4.5 as validation layer (critical quality)

Where Nova 2 Lite shines

Official benchmarks vs Intelligence V3 Tier (Amazon Nova Models Deep Dive)





WHAT THIS MEANS FOR US

RealKIE measures exactly what we needed: structured extraction from documents — in our case, HTML.

The 6 lessons

Part 1 — Efficiency and cost



LESSON 01

Token = Money

Every token is a unit of cost. Measure, compress, and optimize them obsessively. Use CloudWatch to track patterns from day one, not from the first surprise bill.



LESSON 02

Tier Your Models

Not every task needs Sonnet. Route simple extractions to Nova/Haiku. Reserve Sonnet for deep-reasoning workflows.



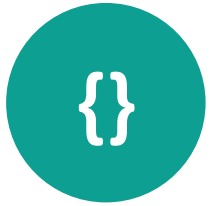
LESSON 03

Cache Aggressively

Semantic cache with DynamoDB/OpenSearch cut our redundant calls by over 50%. Identical products in different formats are still the same product.

The 6 lessons

Part 2 — Reliability and scale



LESSON 04

Schema Contracts Matter

Explicit OpenAPI action schemas eliminate agent hallucinations. Invest time here — it pays off in reliability.



LESSON 05

Rate Limit

InvokeAgentRuntime: 25 TPS per agent/account. We use Step Functions → Lambda → AgentCore with a max 10 concurrent agents parameter.



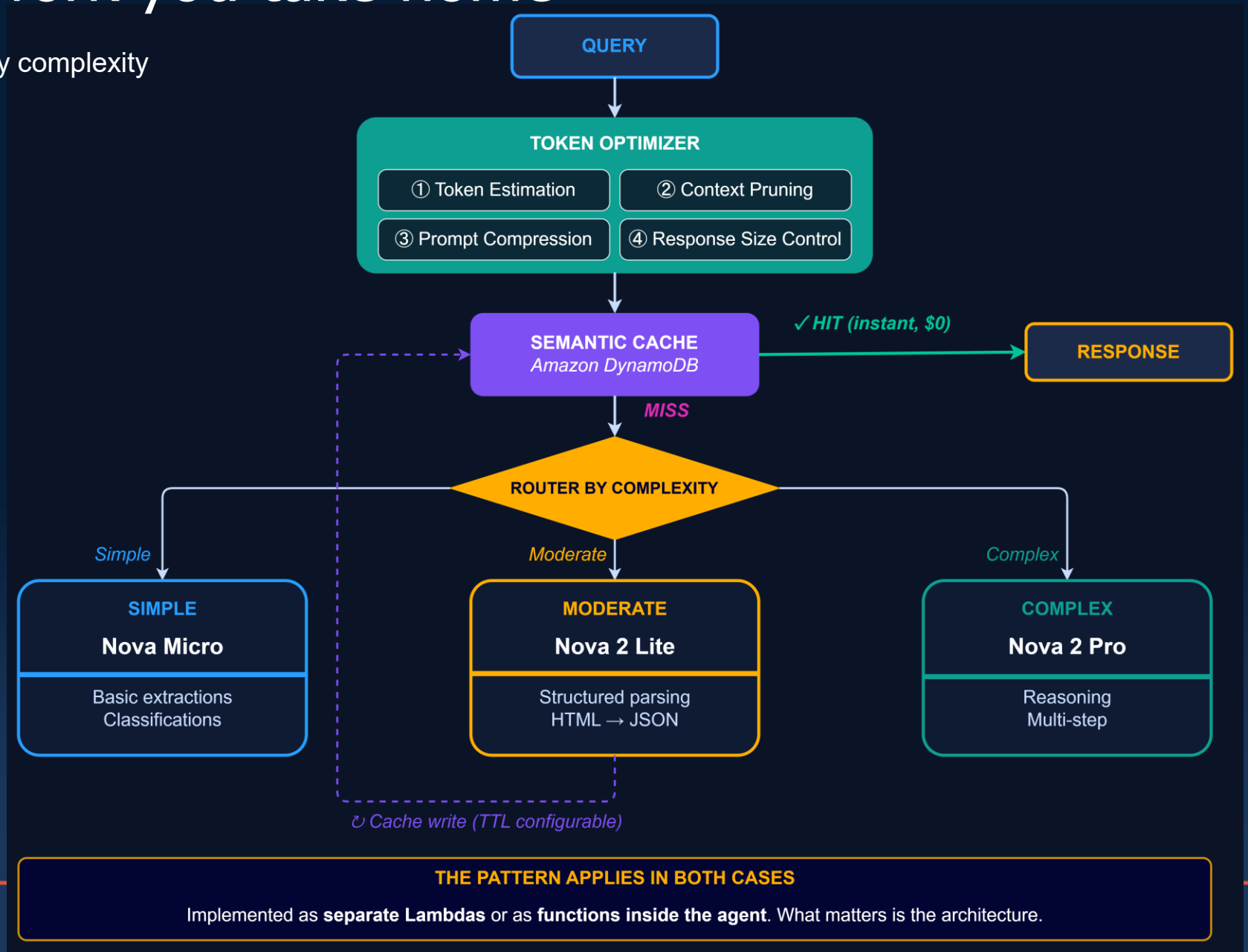
LESSON 06

Start Small, Scale Validated

Build with Nova Micro first. Validate accuracy. Only move up a tier when you have data proving the cost delta is worth it.

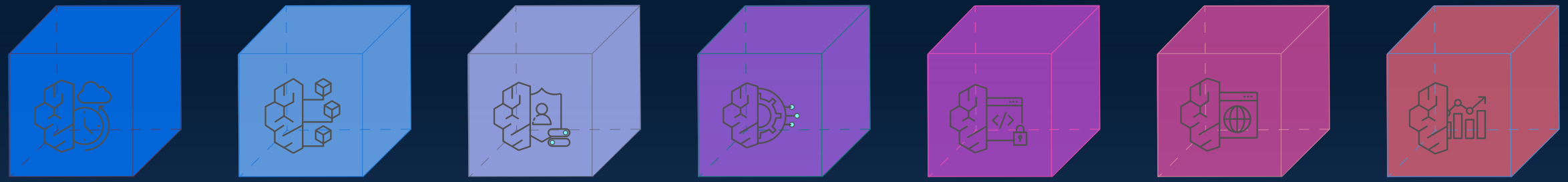
The framework you take home

Semantic cache + routing by complexity



Amazon Bedrock AgentCore

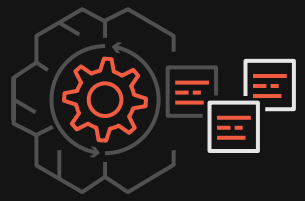
Comprehensive agentic platform: Everything you need for getting agents into production



Runtime | Memory | Identity | Gateway | Code Interpreter | Browser Tool | Observability

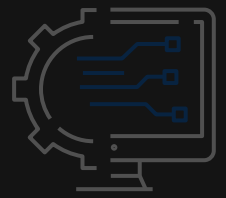
Foundational services for running agents, securely at scale

Enhance with tools & memory



- Memory
- Gateway
- Browser tool
- Code interpreter

Deploy securely at scale



- Runtime
- Identity

Gain operational insights with agentic operations



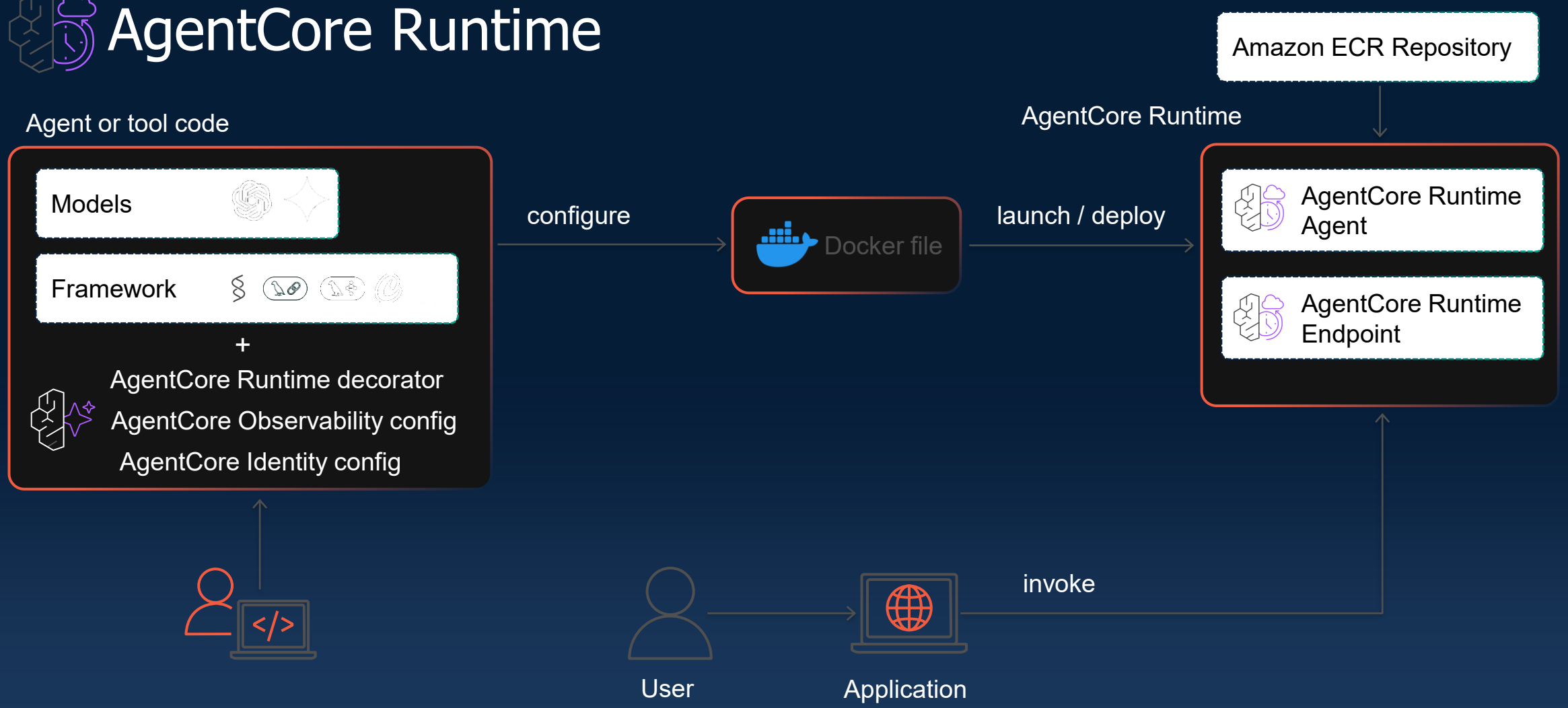
- Observability

Open source

Protocols:
MCP, A2A

Frameworks:
CrewAI, Google ADK, LangGraph, LlamaIndex, OpenAI Agents SDK, Strands Agents

AgentCore Runtime



Demo




$
$3,500
$
$200
$